7 A magnet rotates inside a shaped soft iron core. A coil is wrapped around the iron core as shown in Fig. 7.1. The coil is connected to an oscilloscope.

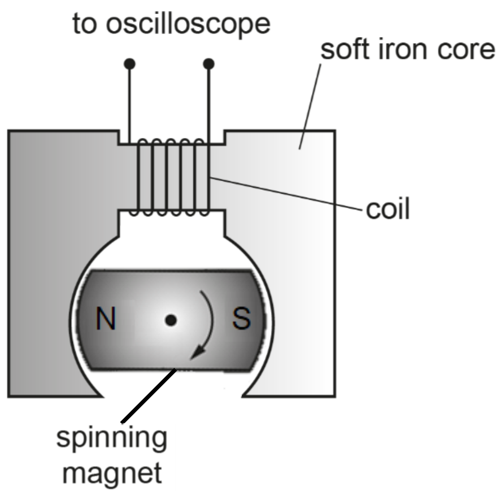


Fig. 7.1

The spinning magnet induces an e.m.f. in the coil.

(a) On Fig 7.2, sketch a graph of the variation of the e.m.f. induced in the coil against time. The variation of the induced magnetic flux linkage in the coil is shown as a dotted line. [1]

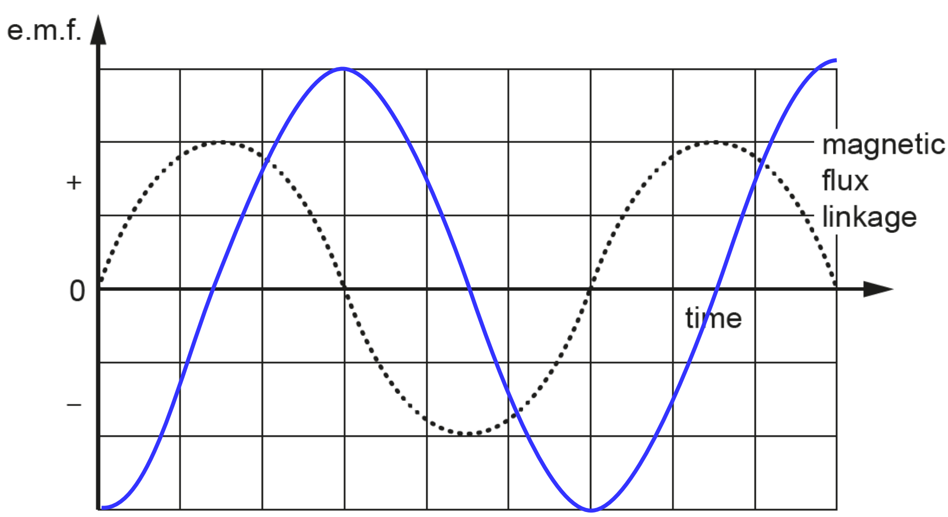


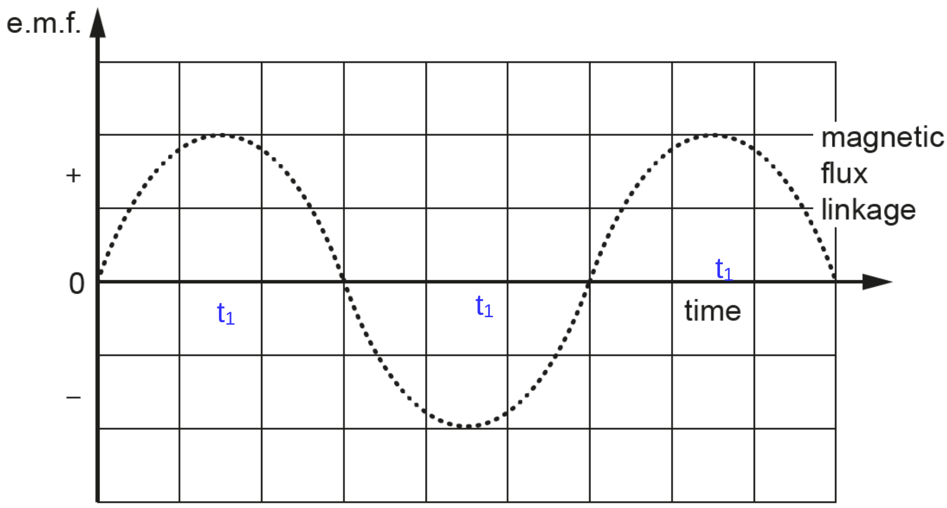
Fig. 7.2

[B1 for cosine or negative cosine curve]

(b) By considering the orientation of the magnet as it spins, explain the variation of the magnetic flux linkage in the coil with time.

1. *Explain maximum and minimum (zero):*
There is maximum flux linkage in the coil is when the magnet is horizontal (with respect to fig 7.1) because the lines of flux from the magnet are parallel with the axis of the coil, and min is when the magnet is vertical, because there are no flux lines through the coil. This variation repeats itself as the magnet continues to rotate.
2. *Explain increase and decrease:*
As the magnet rotates from a horizontal orientation (with respect to fig 7.1) to a vertical orientation, the flux linkage in the coil due to the magnet decreases and as it continues to rotate further to a horizontal orientation, the flux linkage in the coil due to the magnet increases. This variation repeats itself as the magnet continues to spin. [2]
3. *Explain positive and negative:*
When the magnet is in a horizontal orientation (with respect to fig 7.1) with the North pole on the left (with respect to fig 7.1) there is maximum flux linkage in the coil due to the magnet because the lines of flux from the magnet are parallel with the axis of the coil. When the coil has rotated by 180° to being horizontal with the North pole on the right, there is again maximum flux linkage in the coil due to the magnet, but this time it is in the opposite direction, so the magnetic flux linkage at this position has a negative sign compared to when the North pole was on the left. [B1 each for any two features].

(c) At a certain time t_1 the orientation of the spinning magnet is momentarily as shown in Fig 7.1. Mark the time t_1 on the time axis of fig 7.2. [1]



(d) The coil shown in Fig. 5.1 has 150 turns. The maximum induced e.m.f. Vo across the coil is 1.2V when the magnet is rotating at 24 revolutions per second.

Calculate the maximum magnetic flux through the coil.

$\Phi = \Phi_o \sin\theta = \Phi_o \sin\omega t$
Using Faraday's Law
$E = d\Phi/dt = \Phi_o\,\omega\cos\omega t$
Maximum E occurs when $\cos\omega t = 1$
$E_{max} = \Phi_o\,\omega = (2\times\pi\times f)\,\Phi_o$
$1.2 = (2\times\pi\times 24)\,\Phi_o$
$\Phi_o = 5.3\times 10^{-5}$ Wb

maximum flux = ... [2]

[Total: 6]

8 The range of frequencies which can be heard by different people varies, but most can hear